

# Clinical Risk Assessment Report: Patient 92

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 40.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.83x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM\_LGZE.CT), which elevated the patient's absolute risk by +6.3%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Large Zone Emphasis (CT) (GLZLM\_LGZE.CT)** **Value: -0.34 [Bottom 14%] (+6.3%)**  
*Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.*

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)** **Value: 0.86 [Top 11%] (+5.5%)**  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

**Tumor Pixel Correlation (CT) (GLCM\_Correlation.CT)** **Value: 0.54 [Avg (68th %ile)] (+3.5%)**  
*Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: 0.72 [Avg (71th %ile)] (+2.5%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)** **Value: 0.12 [Top 9%] (+2.4%)**  
*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Short Run Low Gray-Level Emphasis (CT) (GLRLM\_SRLGE.CT)** **Value: -0.24 [Bottom 6%] (+2.3%)**  
*Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.*

**High Gray-Level Zone Emphasis (CT) (GLZLM\_HGZE.CT)** **Value: 0.89 [Top 20%] (+1.5%)**  
*Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.*

## 3. Protective / Mitigating Factors (Reducing Risk)

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: -0.08 [Avg (45th %ile)] (-3.4%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

Machine Learning Feature Attribution (SHAP Decision Path)

